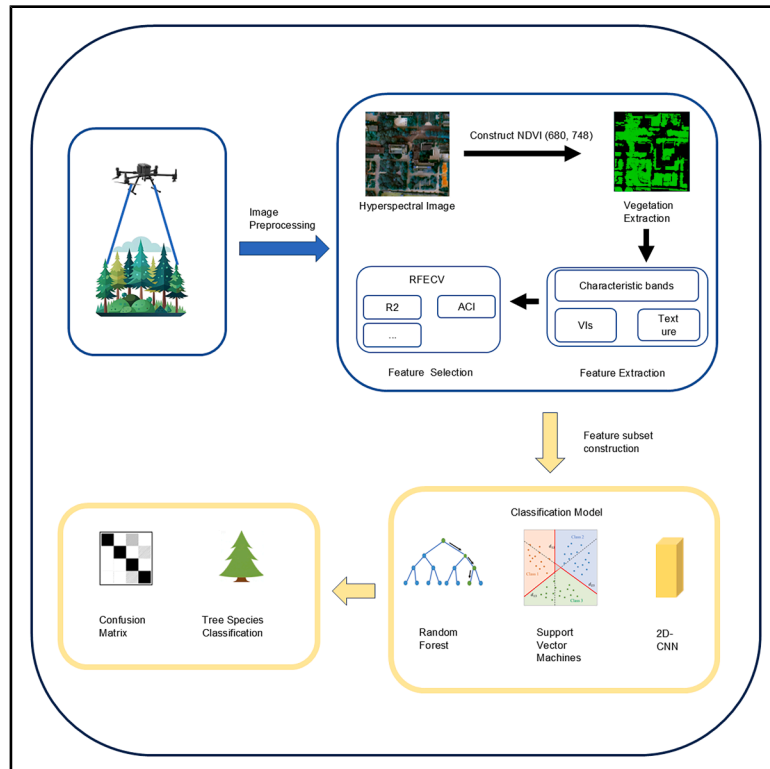


Intelligent classification of dominant tree species in urban forests based on UAV hyperspectral remote sensing images

Graphical abstract



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In brief

Earth sciences; Remote sensing; Forestry; Forest inventory

Highlights

- Developed the NDVI (680,748) vegetation index based on red-edge spectral features
- RFECV method selects key features to maximize tree species identification accuracy
- Evaluating the impact of algorithms and feature combinations on classification
- Provides scalable technical support for urban forest monitoring and urban planning



Article

Intelligent classification of dominant tree species in urban forests based on UAV hyperspectral remote sensing images

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SUMMARY

Urban tree species classification is essential for green space management but is challenged by the fragmented spatial distributions of trees. To address this issue, this study employs a hyperspectral imagery-based classification method to improve the accuracy of urban forest identification. By leveraging the red-edge spectral reflectance characteristics and optimizing band combinations, this study constructs an optimized vegetation index, NDVI (680, 748), which significantly enhances vegetation extraction in urban environments, achieving an overall accuracy of 97.82% and a Kappa coefficient of 0.95. Using this index, recursive feature elimination with cross-validation was applied to select highly discriminative features, which were then integrated into multiple machine learning models. Among these, the random forest model—incorporating characteristic bands, textural features, and vegetation indices—achieved the best test set performance, with an overall accuracy of 86.91% and a Kappa coefficient of 0.85. This method enables high-precision urban forest mapping, facilitating sustainable ecosystem management.

INTRODUCTION

Accelerating urbanization has heightened resident demand for eco-friendly environments, positioning ecological civilization construction as central to urban development. Urban forests—defined as forest ecosystems dominated by trees within and surrounding urban areas, integrated with human-natural environments—constitute a vital foundation for sustainable urban development. They mitigate environmental degradation and enhance urban environments,^{1,2} delivering essential ecosystem services including urban heat island mitigation, noise reduction,^{3,4} carbon sequestration, and mental health protection.⁵ Tree species data provide urban planners and managers with vital information for monitoring, managing, and evaluating urban greening initiatives. As public awareness of urban trees' ecological significance grows, timely and accurate species-level data become imperative for enhancing urban sustainability.^{6,7} However, urban trees within densely populated, artificially landscaped zones exhibit highly fragmented and uneven spatial distributions.⁸ Scientifically and efficiently identifying tree species represents a core

challenge for effective urban planning. Therefore, research on urban forest tree species identification techniques is crucial for advancing urban planning and greening assessment.⁹

Satellite remote sensing, thanks to its efficiency, affordability, and broad spatial reach, has become an indispensable tool for charting the spread of various tree species.¹⁰ Scores of research projects have utilized satellite images to categorize different types of trees. For instance, Zhou et al.¹¹ employed hyperspectral data to identify dominant tree species in Maguan County, while Liang et al.¹² utilized sentinel-1/2 for fruit tree classification in the Tarim Basin. However, satellite imagery acquisition suffers from many problems: high costs, long revisit cycles, and susceptibility to atmospheric interference. Such limitations hinder research and application in urban areas.¹³

Whereas satellite remote sensing faces limitations for refined urban applications, UAV remote sensing offers enhanced flexibility, cost efficiency, simplified processing, and minimal terrain constraints.¹⁴ UAV platforms can be programmed on demand for flight trajectories and sensor parameters, allowing data



Table 1. Specifications of Specim AFX10

Items	Contents
Spectral range	400–1000 nm
Spectral sampling	2.68 nm
Spectral resolution	5.5 nm
Number of spectral bands	224
Number of spatial pixels	1024 px
Field of view	38°
Aperture value	1.7
Dimensions	131 × 152 × 202 mm
Imaging speed	330 fps

acquisition at a specific location, altitude, and time window. UAVs with ultra-high-resolution sensors can achieve sub-decimeter spatial resolution—capturing the feature details necessary for precise monitoring. This new technology provides high-resolution data for local spatiotemporal analyses and for different application scenarios. Consequently, global researchers have leveraged UAV remote sensing for land resource management, environmental monitoring, urban planning, transportation safety, and emergency response.¹⁵ For instance, Zhong et al.¹⁶ identified tree species in complex mixed conifer forests using UAV-derived RGB imagery, whereas Su et al.¹⁷ reconstructed high-resolution urban building models using UAV imagery.

Current hyperspectral remote sensing-based tree species classification commonly employs algorithms including decision trees (DTs), support vector machines (SVMs), random forest (RF), maximum likelihood classification (MLC), K-nearest neighbors (KNNs), and artificial neural networks (ANNs).¹⁸ However, classification accuracy faces constraints from multiple factors, including but not limited to tree species heterogeneity and environmental complexity. Crucially, algorithm performance exhibits significant regional and contextual variability. For instance, Michalowska et al.¹⁹ demonstrated SVM and RF's excellent efficacy in airborne radar-based species identification. Franklin et al.²⁰ achieved accuracies of 60% and 80% using pixel-based and object-oriented methods, respectively, on drone multispectral imagery in mixed forests.

Although hyperspectral imaging has advanced in tree species classification and identification, research applying hyperspectral data to urban forest tree species identification has received less attention.²¹ In order to investigate how different classification algorithms affect the precision in identifying tree species within urban forests, this study selects three classification algorithms and uses the Gongchenqiao Campus of Zhejiang Shuren University as the study area (Figure 1). It focuses on discussing the accuracy of different classification methods in urban forest tree species classification and how different feature combinations influence urban forest tree species classification.

Table 2. Vegetation index calculation formula and main functions

Index	Formulation	Main function
mNDVI ₇₀₅	$(\rho_{750} - \rho_{705}) / (\rho_{750} + \rho_{705} - \rho_{445} \times 2)$	detecting subtle changes in vegetation canopy, forest gaps, and vegetation aging changes
ACI	ρ_{650} / ρ_{550}	estimated ratio of anthocyanin to chlorophyll during leaf development
CI-G	$\rho_{730} / \rho_{530} - 1$	calculate the total chlorophyll content of the leaves.
PSRI	ρ_{695} / ρ_{760}	strongly indicative of the plant stress ratio
R2	ρ_{600} / ρ_{760}	the ratio indicating plant stress
CI-RE	$\rho_{850} / \rho_{730} - 1$	quantifying chlorophyll content in plant leaves
RVSI	$(\rho_{714} + \rho_{752}) / 2 - \rho_{733}$	captured red edges associated with plant stress
mSR ₇₀₅	$(\rho_{750} - \rho_{445}) / (\rho_{750} + \rho_{445})$	can be used for precision agriculture, forest monitoring, vegetation stress detection, and so forth.
SIPI	$(\rho_{800} - \rho_{455}) / (\rho_{800} - \rho_{680})$	detection of canopy stress
PPR	$(\rho_{550} - \rho_{450}) / (\rho_{550} + \rho_{450})$	significantly correlated with leaf nitrogen concentration after flowering
PSSR1	ρ_{800} / ρ_{680}	carotenoid-to-chlorophyll a reflectance ratio vegetation index
PSSR2	ρ_{800} / ρ_{630}	carotenoid-to-chlorophyll b reflectance ratio vegetation index
VOG1	ρ_{740} / ρ_{720}	it is very sensitive to factors such as vegetation canopy moisture content and chlorophyll content.

Table 3. Texture feature calculation formulas and functions

Texture characteristics	Formulation	Main function
Second Moment	$\sum_{i,j=0}^{N-1} P_{ij}^2$	Reflects the uniformity of the grayscale distribution within the image and the coarseness of the texture.
Contrast	$\sum_{k=0}^{N-1} \sum_{i=-1}^k P_{ij} \times k \times k$	Measures the degree of difference between light and dark areas in an image, reflecting the clarity of texture and structural strength.
Correlation	$\sum_{i,j=0}^{N-1} P_{ij} \frac{(i-\mu)(j+\mu)}{\sigma}$	Measures the linear correlation of pixel gray values in a certain direction in an image. It reflects the spatial extension direction or consistency of texture.
Entropy	$\sum_{i,j=0}^{N-1} P_{ij} (-\ln P_{ij})$	Quantifies the randomness or information complexity of a texture; the higher the value, the more chaotic the texture and the richer the details.
Homogeneity	$\sum_{i,j=0}^{N-1} \frac{P_{ij}}{1+(i-j)^2}$	Measure the uniformity of local areas in an image.
Mean	$\sum_{i,j=0}^{N-1} i(P_{ij})$	Calculate the average brightness of pixels to reflect the overall lightness and darkness of the texture.
Dissimilarity	$\sum_{k=0}^{N-1} \sum_{ i-j }^k P_{ij} \times k$	Describes the degree of difference between local pixel values, similar to contrast but calculated differently. Higher values indicate greater differences between adjacent pixels.
Variance	$\sum_{i,j=0}^{N-1} P_{ij} (i-\mu)^2$	Measures the degree of dispersion of pixel values, reflecting the volatility and irregularity of textures.

RESULTS

Experimental environment and process

The experimental configuration included a Lenovo ThinkSystem SR650 server featuring two Intel Xeon Gold 5117 processors, 256 GB of DDR4 memory at 2666 MHz, and an NVIDIA Tesla V100 GPU with 32 GB. The software utilized encompassed Python 3.12, ENVI 6.1, and ArcGIS 10.8. The key parameters of the hyperspectral camera are shown in Table 1.

The experimental process comprised the following steps (Figure 2): (1) Spectral analysis of hyperspectral data to select bands for constructing vegetation indices (Figure 3), with comparison against common indices to evaluate vegetation extraction effectiveness. (2) Feature extraction, including characteristic bands, vegetation indices (Table 2), and texture features (Table

3). (3) Feature selection using RFECV, iteratively removing the least important features based on model contribution (Figure 4). (4) Tree species classification using SVM, RF, and 2D-CNN models, followed by evaluation and comparison on the test data-set (Table 4).

Vegetation extraction and accuracy evaluation

To precisely assess how well vegetation indices extract vegetation from UAV-acquired hyperspectral imagery, this study implemented a threshold-based classification method to generate vegetation images and compare them with commonly used vegetation indices such as EVI,²² GNDVI,²³ NDVI,²⁴ SVI, NDVI (705,750)²⁵ and NDVI (680,748), as shown in Table 5. This was done to determine the optimal classification thresholds for each vegetation index. To fairly compare the

Table 4. Distribution of tree species samples within the study area

Classes	Original training samples (Image block)	Expand training samples (Image block)	Validation samples (Image block)	Test samples (Image block)
Celtis sinensis	5230	20920	1743	1743
Southern magnolias	3000	12000	1000	1000
Osmanthus tree	5166	20664	1722	1722
Pine tree	3632	14528	1210	1210
Loquat tree	3675	14700	1225	1225
Chinese tallow tree	5358	21432	1786	1786
Camphor tree	7948	31792	2649	2649

Table 5. Comparison of vegetation indices

Index	Formulation
EVI	$2.5 \times (\rho_{800} - \rho_{680}) / (\rho_{800} + 6 \times \rho_{680} - 7.5 \times \rho_{450} + 1)$
GNDVI	$(\rho_{801} - \rho_{550}) / (\rho_{801} + \rho_{550})$
NDVI	$(\rho_{860} - \rho_{670}) / (\rho_{860} + \rho_{670})$
SVI	$(\rho_{820} - \rho_{670}) / (\rho_{820} + \rho_{670}) \times \rho_{820}$
NDVI (705,750)	$(\rho_{750} - \rho_{705}) / (\rho_{750} + \rho_{705})$
NDVI (680,748)	$(\rho_{748} - \rho_{680}) / (\rho_{748} + \rho_{680})$

performance of various vegetation indices, we iterated through all threshold values within the interval [0, 0.8] with a step size of 0.01, calculating the classification results for each threshold. Ultimately, the optimal threshold for each vegetation index was selected based on a comprehensive evaluation metric that equally weights (50% each) the overall accuracy (OA) and the Kappa coefficient, as shown in Table 6. In the binary vegetation images, green pixels denoted vegetation areas, while non-vegetation areas were depicted in black to visually represent the outcomes of vegetation information extraction. Furthermore, four distinct target areas within the study region were chosen to exemplify various land cover types for comparative analysis, encompassing shaded areas, building areas, Roof, basketball court, and lawn and shrub areas. The results of vegetation information extraction under different vegetation indices are shown in Figure 5.

The binary images of each vegetation index in Figure 5 provide a visual reference for assessing the accuracy of vegetation extraction. In Figure 5A, except for NDVI (680, 748), all other vegetation indices classify the basketball court as vegetation to varying degrees. In Figure 5B, SVI and EVI classify the white and orange-red buildings as vegetation, while GNDVI only classifies the orange-red building as vegetation. In Figure 5C, none of the vegetation indices misclassifies shadows as vegetation; however, EVI misclassifies the white ground as vegetation, and SVI performs poorly in extracting vegetation in the shadowed areas. In Figure 5E and SVI fails to extract vegetation in the shadowed areas, while EVI misclassifies the white ground and some buildings as vegetation.

In order to quantitatively assess the accuracy of each vegetation index, this study used the confusion matrix method to

Table 6. Optimal vegetation index thresholds

Index	Optimal threshold
EVI	0.19
GNDVI	0.42
NDVI	0.46
SVI	0.10
NDVI (705,750)	0.40
NDVI (680,748)	0.40

evaluate their accuracy in the experimental area, as shown in Table 7.

As shown in Table 7, all vegetation indices demonstrate varying degrees of vegetation extraction capability. Among these indices, NDVI (680, 748) achieved the highest OA and Kappa coefficient. GNDVI, NDVI, and NDVI (705, 750) also exhibited good vegetation extraction performance. However, EVI showed low non-vegetation producer accuracy and vegetation user accuracy, indicating that it misclassifies a large number of non-vegetated areas as vegetation. However, EVI exhibits low non-vegetation producer accuracy, indicating that a significant number of non-vegetation pixels are misclassified as vegetation. Simultaneously, its vegetation user accuracy is also low, suggesting that pixels classified as vegetation actually contain a considerable amount of non-vegetation objects. Overall, this reflects a tendency toward the over-extraction of vegetation. Conversely, SVI exhibits low producer accuracy for vegetation, reflecting numerous omissions of vegetated pixels. Its non-vegetation user accuracy is also low, meaning pixels classified as non-vegetation contain considerable vegetation contamination. Overall, SVI demonstrates insufficient vegetation extraction.

Tree species classification experiment

In this study, three different sets of feature variables were extracted to evaluate their effectiveness in tree species classification tasks. To systematically evaluate the impact of different features and classifiers on tree species classification performance, four schemes were designed: ALL bands, characteristic bands, characteristic bands + vegetation indices, and characteristic bands + vegetation indices + texture features. These were

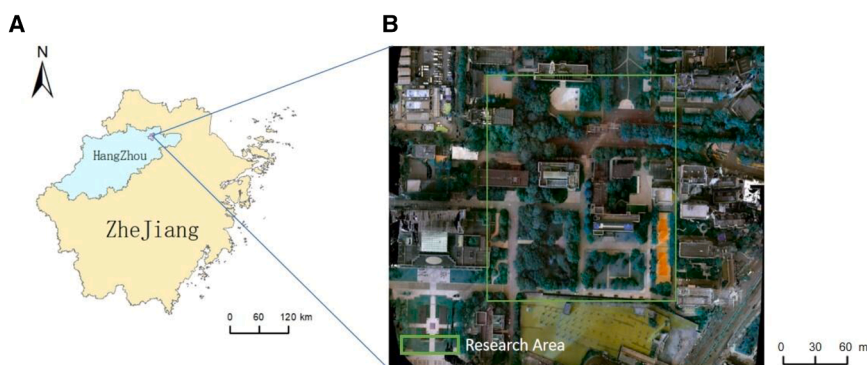
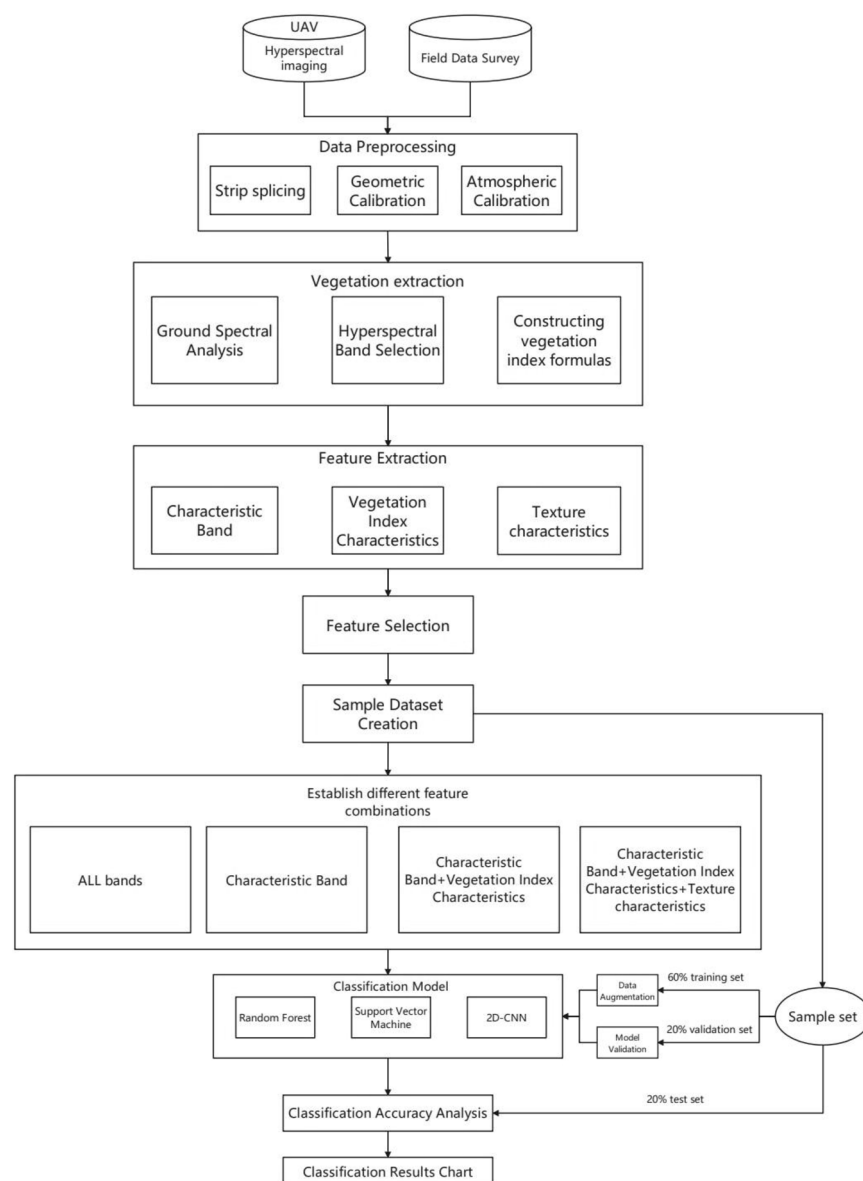


Figure 1. Research area

(A) Indicates the location of the study area. The scale bar has a total length of 120 km and is divided into two equal segments, each representing 60 km. (B) Digital orthophotos of the study area. The scale bar has a total length of 60 m and is divided into two segments of 30 m each. The spatial resolution is 0.1 m per pixel.

**Figure 2. Technology roadmap**

more, the feature set reduced via MNF demonstrated higher classification potential than the full-band approach, indicating that feature selection and combination play a crucial role in enhancing model performance. Detailed accuracy results for each approach are presented in [Table 9](#), while hyperspectral image-based tree species classification results are shown in [Figure 6](#).

The influence of feature combinations on the classification accuracy of urban tree species

When classification is based solely on characteristic bands, analyzing the classification results of the two types of classifiers shows that the RF classifier achieved the best performance on the validation set, with an OA of 83.27% and a Kappa coefficient of 0.80. As shown in Experiment Result S4, under the RF classifier, the tree species with the lowest user accuracy is the Camphor tree, with a user accuracy of only 65.97%, followed by the Chinese tallow tree and osmanthus tree, with user accuracies of 80.32% and 73.37%, respectively. This indicates that there is a significant misclassification issue for Camphor tree, Chinese tallow tree, and osmanthus trees in the RF classifier. Additionally, in experimental result S3, the Camphor tree has the lowest producer accuracy, at only 53.28%, indicating severe misclassification. The potential cause of misclassification arises from the similar spectral characteristics shared

combined with RF, SVM classification, and 2D-CNN, as shown in [Table 8](#).

Analysis of tree species classification accuracy

This study systematically compared the performance of nine identification schemes on the validation set. Results indicate that the scheme combining MNF-transformed features, vegetation indices, and texture features fed into a RF classifier achieved the optimal classification performance, with an OA of 86.91%. In contrast, when using only MNF-transformed features, the classification accuracies of the RF and SVM were 83.27% and 80.17%, respectively. When directly utilizing all original spectral bands, the accuracies of both methods decreased to 74.62% and 75.03%. Overall, the multi-feature fusion approach based on RF significantly outperformed single-feature or full-band methods. Further-

by these tree species, which makes them difficult to distinguish from other categories. Additionally, class imbalance is present, as the number of camphor tree samples after data augmentation exceeds that of other categories, resulting in a significant number of misclassified samples. In contrast, the classification accuracy of *Celtis sinensis*, southern magnolias, and pine trees is relatively good, indicating that the classifiers have good recognition of these tree species.

In summary, spectral characteristics play an important role in the classification of dominant tree species. However, when tree species have similar spectral characteristics, it leads to severe misclassification and omission. This similarity causes many samples to be incorrectly classified or omitted.

The addition of vegetation indices significantly impacts classification accuracy. The overall classification accuracy was

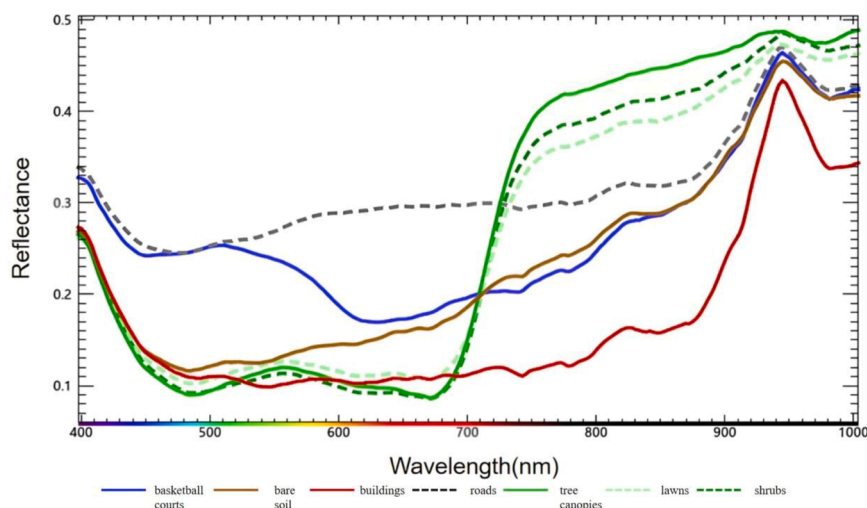


Figure 3. Spectral reflectance curves of typical landforms

significantly improved for both classification methods. Specifically, RF's overall classification accuracy increased by 2.39% (from 83.27% to 85.66%), and the Kappa coefficient improved by 0.03 (from 0.80 to 0.83); SVM's overall classification accuracy improved by 0.96% (from 80.17% to 81.13%), and the Kappa

coefficient increased by 0.01 (from 0.78 to 0.79). The accuracy of most tree species improved to varying degrees, while the accuracy of some species decreased. For instance, in the RF classifier, the producer accuracy of the Chinese tallow tree decreased by 0.56%. This may be because plants grow vigorously, and most tree species appear dark green, resulting in high chlorophyll content. At this stage, the vegetation index proves exceptionally responsive to variations in vegetation. However, under the condition of full canopy cover, the canopy closure of trees is high, and factors such as tree shading, shadows, and forest gaps may also affect the performance of leaves in vegetation indices. These factors may lead to increased variability in vegetation indices and significant differences in mean values among tree species.

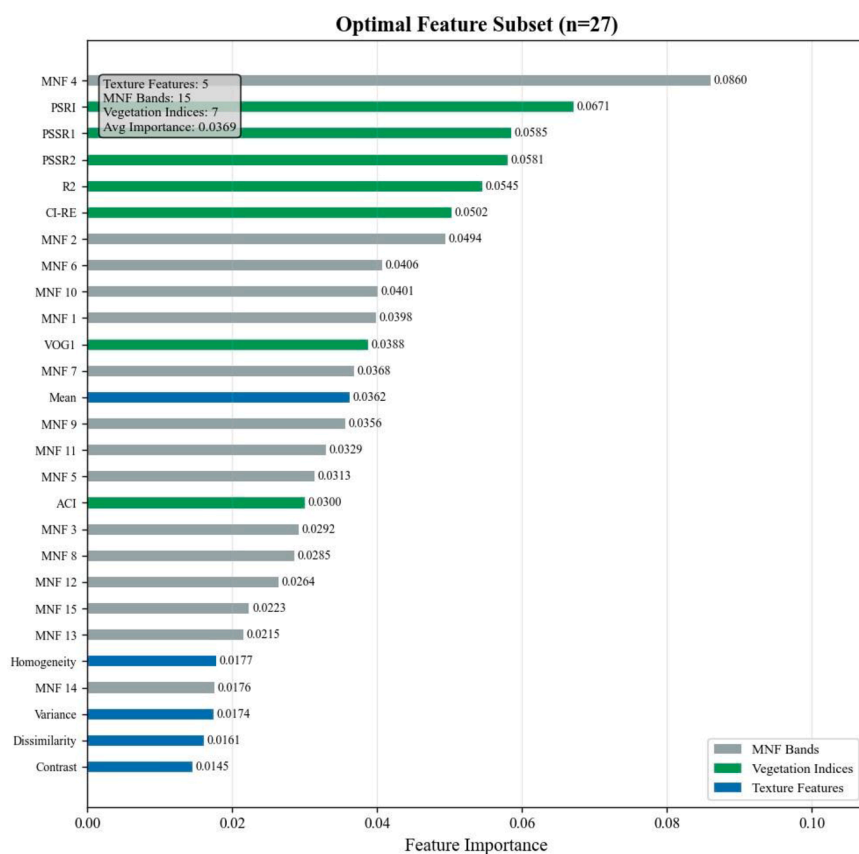


Figure 4. Optimal feature subset and importance scores

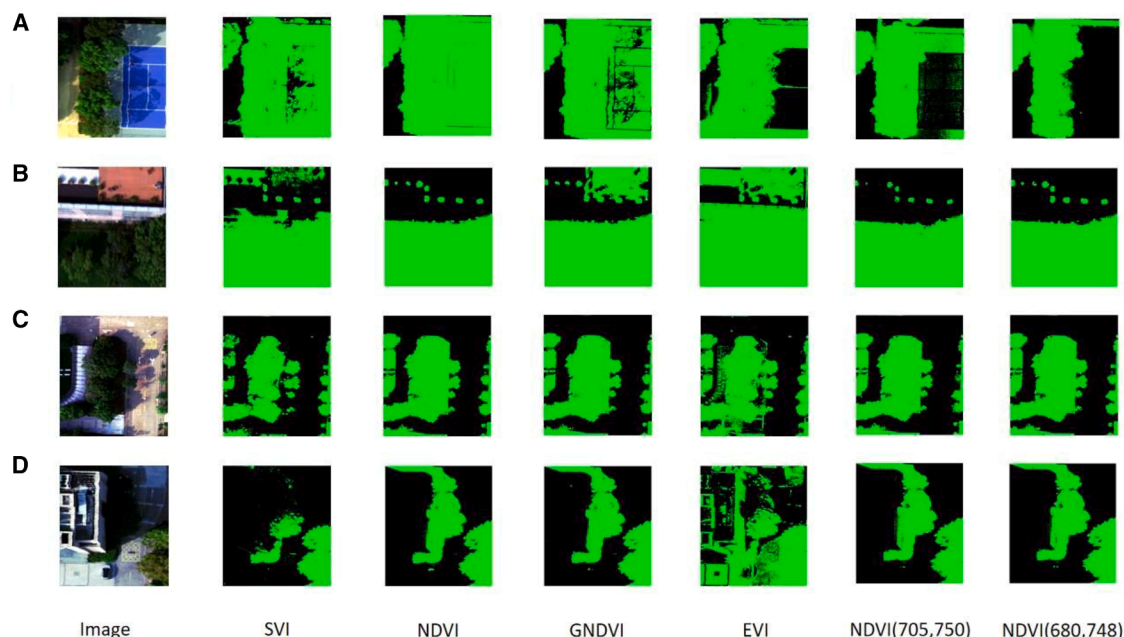


Figure 5. Results of vegetation information extraction under different vegetation indices

The addition of texture features significantly affects classification accuracy. The overall classification accuracy of both classification models changed significantly. Specifically, RF's overall classification accuracy improved by 1.25%, with the Kappa coefficient increasing by 0.02, while SVM's overall classification accuracy improved by 3.18%, with the Kappa coefficient also increasing by 0.03.

As shown in Table 9, the OA improved significantly after adding texture features, but the accuracy in classifying certain tree species has declined. For example, in the RF model, the user accuracy and producer accuracy of the loquat tree both decreased to some extent. In the SVM model, the user accuracy of the *Celtis sinensis* decreased by 3.79%. In summary, texture features have important impact on urban forest tree species identification. In urban environments, the introduction of texture features plays a crucial role in tree species classification models. The incorporation of texture features significantly alters the overall classification accuracy of urban forest tree species. However, different tree species exhibit varying responses to texture features. This may be due to differences in canopy size among tree species, while this study uses a fixed 5×5 window to extract texture fea-

tures. For example, the canopy of the *Celtis sinensis* is typically broad and naturally rounded, with uneven leaf density and a large canopy size, making its edge structure more complex. Under the 5×5 fixed window, the extracted texture features do not fully represent the overall canopy structure. In contrast, the loquat tree has a smaller canopy with large, leathery, and smooth leaves, which causes the window to extend beyond the canopy boundary, leading to the extraction of excessive background information around the canopy.

The S9 (2D-CNN with characteristic bands only) scheme does not incorporate texture or vegetation index features in classification, relying solely on the spectral bands selected from the raw hyperspectral data. Compared to the S8 scheme, which uses multiple features, the OA decreased by 0.36%, and the Kappa coefficient dropped by 0.01. This suggests that the combination of multiple features compensates for the limitations of the machine learning model's inherent learning capability, while the 2D-CNN model does not exhibit its advantages when the feature set is limited. This indicates that in tree species classification tasks, under identical sample sets, model performance largely depends on the feature information provided.

Table 7. Evaluation of vegetation index accuracy

Index	PA_Veg(%)	PA_Non-Veg(%)	UA_Veg(%)	UA_Non-Veg(%)	OA(%)	Kappa
EVI	90.30	74.79	78.46	88.35	82.61	0.65
GNDVI	97.05	96.25	96.34	96.98	96.65	0.93
NDVI	98.40	95.14	95.37	98.31	96.78	0.93
SVI	83.28	92.98	92.34	84.54	88.08	0.76
NDVI (705,750)	95.63	97.51	97.51	95.64	96.56	0.93
NDVI (680,748)	97.73	97.93	97.96	97.70	97.82	0.95

Table 8. Tree species identification scheme

Schemes	Classification Features	Characteristic bands	VIs	Textures	Total Features	Classification Model
S1	ALL bands	224	–	–	224	SVM
S2	ALL bands	224	–	–	224	RF
S3	characteristic bands	15	–	–	15	SVM
S4	characteristic bands	15	–	–	15	RF
S5	characteristic bands + VIs	15	7	–	22	SVM
S6	characteristic bands + VIs	15	7	–	22	RF
S7	ALL	15	7	5	27	SVM
S8	ALL	15	7	5	27	RF
S9	characteristic bands	15	–	–	15	2D-CNN

This study focused on the Gongchenqiao campus of Zhejiang Shuren University as the study area. An optimized vegetation index was used, significantly improving vegetation extraction accuracy. Based on this, all bands, spectral bands, vegetation indices, and texture features were input into the RF, SVM, and 2D-CNN models for tree species classification. The main conclusions are as follows.

- (1) For classification models, characteristic bands play a crucial role in tree species classification. The addition of vegetation indices and texture features can improve the overall classification accuracy to a certain extent.
- (2) For the seven dominant tree species, this study established an optimal identification scheme through systematic feature combination and classifier comparison. Four progressive feature sets were constructed: all bands, characteristic bands, characteristic bands combined with vegetation indices, and a multi-feature set inte-

grating characteristic bands, vegetation indices, and texture features. These were input into RF, SVM, and 2D-CNN classifiers for training and testing. Results on the test set show that the RF model using multi-feature inputs achieved the best performance, with OA surpassing both SVM and the 2D-CNN model based solely on characteristic bands. The RF model combining “characteristic bands + vegetation indices + texture features” yielded the best classification results, with an OA of 86.91% and a Kappa coefficient of 0.85, enabling high-precision mapping of dominant tree species distribution. This indicates that, given identical sample sets, model performance largely depends on the feature information provided.

- (3) This study developed an optimized vegetation index, NDVI (680, 748), which demonstrated better extraction accuracy compared to other conventional vegetation indices in this study area.

Table 9. Tree species identification results based on feature combinations in the hyperspectral image

Class	Celtis sinensis	Southern magnolias	Osmanthus tree	Pine tree	Loquat tree	Chinese tallow tree	Camphor tree	OA	Kappa
S1-UA	92.73	91.75	84.57	88.33	85.42	60.51	40.26	75.03	0.67
S1-PA	66.01	34.23	46.74	73.57	46.48	55.43	95.08	–	–
S2-UA	92.05	89.55	84.77	88.64	84.99	62.64	39.52	74.62	0.66
S2-PA	67.09	37.30	44.26	72.72	34.30	54.88	94.52	–	–
S3-UA	91.50	82.57	60.96	80.53	67.86	79.10	67.27	80.17	0.78
S3-PA	91.50	90.00	69.50	91.00	66.50	70.00	53.28	–	–
S4-UA	84.93	93.15	73.37	96.48	90.05	80.32	65.97	83.27	0.80
S4-PA	95.89	79.33	77.35	79.34	70.20	78.82	73.02	–	–
S5-UA	91.84	84.69	63.85	87.20	63.16	75.12	69.23	81.13	0.79
S5-PA	90.00	88.50	68.00	92.00	66.00	75.50	56.41	–	–
S6-UA	88.06	92.52	75.43	97.73	93.25	81.93	71.56	85.66	0.83
S6-PA	96.56	84.50	81.12	85.26	77.01	78.26	76.02	–	–
S7-UA	88.05	94.04	73.33	97.45	89.92	81.54	68.49	84.31	0.82
S7-PA	95.12	84.17	77.44	84.16	74.01	76.21	74.95	–	–
S8-UA	89.76	96.92	77.48	98.76	89.29	83.67	73.73	86.91	0.85
S8-PA	97.23	89.17	86.25	87.60	74.83	81.72	75.72	–	–
S9-UA	90.50	89.03	72.85	92.67	82.85	75.72	86.72	86.55	0.84
S9-PA	96.83	92.00	81.83	93.50	85.33	70.17	73.90	–	–

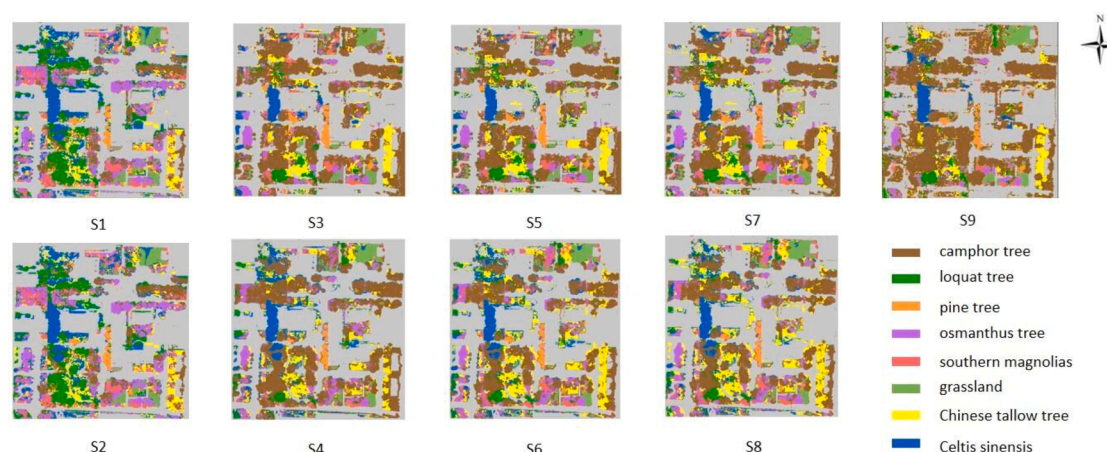


Figure 6. Hyperspectral image tree species identification results for various feature combinations

DISCUSSION

This study developed an optimized NDVI (680, 748) that performs well in vegetation extraction, leveraging the rich spectral information of hyperspectral data. Unlike traditional indices based on broad spectral ranges, hyperspectral data enable systematic searching within the red-edge region (680–750 nm) to pinpoint the optimal dual-band combination (680 nm and 748 nm). This combination exploits the steep reflectance increase in the red-edge region, enhancing spectral separability between vegetation and non-vegetation. Experimental results show that this index effectively suppresses spectral interference from typical urban features, such as basketball courts and colored rooftops, significantly reducing misclassification rates and demonstrating its robustness in complex urban environments.

This study investigated key factors in urban forest tree species classification, including characteristic bands, texture features, vegetation indices, classification methods, and feature selection. Experimental results show that the RF algorithm achieved the best performance on the test set, attaining an OA of 86.91% and a Kappa coefficient of 0.85 when integrating characteristic bands, texture features, and vegetation indices. The proposed high-precision classification method enables detailed mapping of urban tree spatial distribution and supports monitoring of tree growth, stress responses, and health status, providing critical data and decision-making support for urban green space planning, ecological function assessment, pest and disease early warning, and sustainable maintenance management. Additionally, the classification accuracy of each tree species was analyzed, emphasizing the critical role of characteristic bands in tree species classification and the use of an optimized vegetation index for vegetation extraction. Overall, the addition of vegetation index features improved the OA of tree species identification, but for some tree species, accuracy may decrease due to factors such as canopy closure and shading. Additionally, the impact of texture features on tree species identification was explored. Adding texture features generally improved the OA of tree

species identification, but for some tree species, accuracy decreased.

Limitations of the study

This study used single-temporal hyperspectral data for tree species identification, which limits the generalizability of the findings. For example, Liu²⁶ demonstrated that multi-temporal data generally yield superior identification results compared to single-temporal data. Further investigation is needed to examine how spectral bands of the same tree species vary under different environmental conditions and how species perform across temporal and spatial contexts. Additionally, this study relied solely on UAV hyperspectral data without incorporating multi-source remote sensing data. Multi-source data fusion, such as combining hyperspectral with LiDAR, can provide complementary structural information that aids tree species identification. For instance, Lin et al.²⁷ successfully classified tree species across different elevations using multi-source data. Future research should therefore integrate multi-source remote sensing data to improve classification accuracy and robustness.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Shiping Ye (ysp@zjzsu.edu.cn).

Materials availability

This study did not generate new unique compounds.

Data and code availability

- Datasets have been deposited in (Mendeley Data) and are accessible at (<https://data.mendeley.com/datasets/zsfshdgvz/1>).
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.Y., S.S., and V.G.; methodology, B.L., S.Y., Z.B., C.Y., and Y.X.; software, B.L.; validation, B.L., S.Y., Z.B., C.Y., Y.X., S.S., and V.G.; formal analysis, B.L., S.Y., Z.B., C.Y., K.Z., Y.X., S.S., and V.G.; investigation, B.L., S.Y., Z.B., C.Y., K.Z., and Y.X.; resources, S.Y.; data curation, Z.B., C.Y., Y.X., and K.Z.; writing – original draft, B.L.; writing – review and editing, S.Y.; visualization, B.L., Z.B., C.Y., Y.X., and K.Z.; supervision, S.Y., S.S., and V.G.; project administration, S.Y.; funding acquisition, S.Y. All authors have read and agreed to the published version of the manuscript.

DECLARATION OF INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Hyperspectral Image	This paper ; Mendeley Data	https://doi.org/10.17632/zsfshdgvz.1 om/preview/zsfshdgvz?a=1cdceba-4bbb-4f38-9b80-faa245475f67
Software and algorithms		
The Environment for Visualizing Images (ENVI)	Exelis Visual Information Solutions	www.nv5geospatialsoftware.com
Python, version 3.12	Python Software Foundation	https://www.python.org
Google Earth Engine	Google/Earth Engine Team	https://earthengine.google.com/
ArcGIS 10.8	Esri	www.esri.com

METHOD DETAILS

Overall methodology

This study developed a workflow for classifying dominant tree species in urban forests using high-resolution hyperspectral imagery acquired by an unmanned aerial vehicle (UAV). The methodology, illustrated in Figure 2, comprises six main steps: (1) UAV data acquisition and preprocessing; (2) spectral analysis of typical land covers and construction of an optimized normalized difference vegetation index using red-edge bands for accurate vegetation extraction; (3) feature extraction from the hyperspectral imagery, including characteristic bands selected via Minimum Noise Fraction (MNF) transformation, vegetation index features, and texture features derived from the Gray-Level Co-occurrence Matrix (GLCM); (4) feature selection using Recursive Feature Elimination with Cross-Validation (RFECV) to identify the most discriminative feature subset; (5) construction of multiple classification schemes with different feature combinations (e.g., characteristic bands only, characteristic bands plus vegetation indices, and multi-feature sets incorporating texture); and (6) tree species classification and accuracy assessment using three machine learning models: Random Forest (RF), Support Vector Machine (SVM), and 2D-CNN. Model performance was evaluated using overall accuracy, Kappa coefficient, producer accuracy, and user accuracy based on validation and test datasets. The ultimate goal of this study is to establish a high-precision mapping framework for urban forest tree species and to systematically investigate the influence of feature sets and classification algorithms on identification accuracy in complex urban environments.

Study areas and datasets

The research was carried out at the Gongchenqiao Campus of Zhejiang Shuren University (30°N, 120°E), located in the Gongshu District of Hangzhou, Zhejiang Province, China (see Figure 1). Positioned in an area characterized by a subtropical monsoon climate, Hangzhou experiences distinct seasonal rainfall patterns, with an average annual temperature of 13–20°C and an annual average precipitation of approximately 1,400 millimeters, providing ample water for vegetation.²⁸ The campus has a variety of land cover types, high vegetative cover, and a rich urban forest tree species. This extensive vegetation cover not only enhances landscape aesthetics but also improves air quality, climate conditions, and mitigates the urban heat island effect.²⁹ Consequently, the site's varied land cover, high vegetation density, and rich tree species diversity make it an ideal location for this study.

This study employed a DJI Matrix 300 RTK UAV for data collection. The UAV was equipped with an RTK module to ensure high-precision positioning. A Specim AFX10 camera was used to collect high-resolution hyperspectral images, capturing images in 224 bands with a spatial resolution of 0.03 meters. Key parameters are shown in Table 1.

The flight plan was formulated using DJI Pilot 2, and the full operation was executed in flight path mode. The equidistant interval photography mode was automatically activated to ensure efficient image acquisition. In this study, Raw hyperspectral data were processed in ENVI 6.1 through radiometric, geometric, and atmospheric correction, followed by mosaicking and cropping. Spectral smoothing was applied to reduce noise. The data were then resampled to a spatial resolution of 0.10 m using the nearest neighbor method to reduce processing complexity while preserving data quality.³⁰

Field data survey

Zhejiang Shuren University's Gongchenqiao Campus boasts a rich variety of tree species, with evergreen broadleaf forests predominating. These include loquat trees, metasequoias, podocarpus, Chinese fringe flowers (Loropetalum), mimosa trees (Albizia

julibrissin), southern magnolias (*Magnolia grandiflora*), osmanthus trees, pines, Chinese tallow trees, camphor trees, camellias, maples, and purple-leaf plum trees. However, some tree species have low numbers and small crown spreads. Therefore, after comprehensively considering the quantity of each tree species and the extent of their crown spreads, the dominant tree species for the urban forest were determined to include: *Celtis sinensis* (*Celtis sinensis* Pers), southern magnolias (*Magnolia grandiflora*), osmanthus trees (*Osmanthus fragrans*), pine trees (*Pinus*), loquat trees (*Eriobotrya japonica*), Chinese tallow trees (*Sapium sebiferum*), and camphor trees (*Cinnamomum camphora*).

Vegetation extraction

To eliminate spectral interference from non-vegetation areas, this study analyzed the spectral reflectance of typical land cover types in order to construct effective vegetation indices for vegetation extraction.³¹ In the study, Regions of Interest (ROIs) were meticulously selected within the high-resolution hyperspectral imagery captured by the UAV. Subsequently, the spectral reflectance curves associated with typical land cover types were examined in detail. The range of land cover types under examination included a diversity of plant communities like shrubs, tree canopies and lawns, along with non-vegetation types such as roads, basketball courts, buildings and bare soil, as illustrated in Figure 3.

As shown in Figure 3, vegetation exhibits spectral curves distinct from those of non-vegetation. All vegetation types—tree canopies, shrubs, and lawns—display a characteristic “two valleys and one peak” pattern. The first absorption valley occurs near 490 nm, with low reflectance and gradual variation between 400–500 nm. Reflectance then increases to a peak around 550 nm, attributed to chlorophyll reflection. Following this, reflectance declines to the second absorption valley (the “red valley”) between 670–680 nm, in which strong chlorophyll absorption results in minimal reflectance. Between 680 and 750 nm, reflectance rises sharply—a phenomenon known as the red-edge effect—with vegetation exhibiting significantly higher reflectance variability than non-vegetation in this region, enabling clear differentiation between them.³² Specifically, typical vegetation and non-vegetation can be distinguished by reflectance contrasts at 680 nm to 750 nm. Therefore, to systematically evaluate the performance of red edge band combinations in urban vegetation identification, this study constructed a series of normalized vegetation indices within the red edge range. To determine the optimal combination and enhance efficiency, we adopted a systematic search strategy: After sequentially numbering the bands within the red edge (approximately 680 nm to 750 nm), we arranged the bands in wavelength order and sampled every other band, thereby constructing a set comprising 91 candidate bands. Subsequently, we exhaustively evaluated all possible two-band combinations within this set to compute the corresponding vegetation indices. To fairly compare the performance of various vegetation indices, we iterated through all threshold values within the interval [0, 0.8] with a step size of 0.01, calculating the classification results for each threshold. Based on manually annotated ground truth maps, we computed overall accuracy and Kappa coefficients via confusion matrices to determine precision and optimal thresholds. The optimal band combination was selected by comprehensively comparing the performance of all indices at their respective optimal thresholds, with overall accuracy and the Kappa coefficient each assigned an equal weight of 50% in the evaluation. Experimental results indicate that the combination of 680 nm and 748 nm performs best at a threshold of 0.4, achieving an overall accuracy (OA) of 97.82% and a Kappa coefficient of 0.95. The formula is as follows:

$$NDVI(680, 748) = \frac{\rho_{748} - \rho_{680}}{\rho_{748} + \rho_{680}} \quad (\text{Equation 1})$$

In the formula, ρ_{748} represents the reflectance at a wavelength of 748 nm, and ρ_{680} represents the reflectance at a wavelength of 680 nm.

After calculating the vegetation index, the image is binarized to generate a vegetation mask, marking the vegetation area as foreground (value = 1) whereas all other sections as background (value = 0). The binarization formula is:

$$M(x, y) = \begin{cases} 1, & NDVI(680, 748) \geq \tau \\ 0, & NDVI(680, 748) < \tau \end{cases} \quad (\text{Equation 2})$$

$M(x, y)$ denotes the binarized vegetation mask image, whereas τ indicates the threshold that distinguishes vegetation from the background.

Characteristic bands filtering

Raw hyperspectral images contain numerous bands, making analysis of all bands computationally intensive and introducing redundant information that may compromise classification accuracy.³³ Characteristic band selection aims to reduce dimensionality while maintaining or enhancing classification performance. The Minimum Noise Fraction (MNF) transformation concentrates essential image information into the first few components.³⁴ Its primary functions include determining the intrinsic dimensionality of the image data, separating noise from meaningful signals, and reducing data complexity for subsequent processing. In hyperspectral analysis, MNF serves as an effective technique for extracting principal features and suppressing noise. The following section outlines the fundamental principles of the Minimum Noise Fraction transformation in hyperspectral processing.

- (1) Constructing the covariance matrix: The core of MNF is to process data by calculating the covariance matrix of hyperspectral data. The covariance matrix contains the statistical relationships between data bands.

$$X = M \times N \quad (\text{Equation 3})$$

$$C = \frac{XX^T}{n - 1} \quad (\text{Equation 4})$$

In this context, X represents hyperspectral data, M represents the number of bands, N represents the number of pixels, C represents the covariance matrix, and T represents the transpose operation.

- (2) Eigenvalue decomposition: Eigenvalue decomposition is performed on the covariance matrix to obtain eigenvalues, which represent the amount of information, and corresponding eigenvectors, which indicate the direction of each eigenvalue.
- (3) Minimum noise separation: After eigenvalue decomposition, sort the eigenvalues and select the smallest k eigenvalues (typically $k \leq m$). The eigenvectors corresponding to these k eigenvalues form the MNF transformation matrix. This transformation matrix projects the hyperspectral data into a new space where the variance of noise is minimized.
- (4) Projection and decorrelation: the original hyperspectral data is projected onto the new space defined by the MNF transformation matrix, yielding the data after minimum noise separation. In this space, spectral bands become more independent, reducing noise impact and highlighting the primary information in the data.

This study employed the Minimum Noise Fraction (MNF) transformation to reduce the dimensionality of hyperspectral data, retaining the top 30 MNF bands with the largest eigenvalues. These bands collectively contributed approximately 96.6% of the total variance, meaning that while noise was suppressed to the greatest extent possible, the vast majority of the effective signal in the original data was preserved.

Vegetation index characteristics

Hyperspectral data enable the construction of diverse vegetation indices through band differential or ratio transformations. These indices reflect variations in leaf area, canopy structure, chlorophyll content, and moisture across tree species, demonstrating discriminative capacity for species identification.³⁵ As shown in Table 2, 14 vegetation indices commonly used for tree species classification in the literature were selected, including the modified normalized difference vegetation index (mNDVI₇₀₅),³⁶ Anthocyanin Content Index (ACI),³⁷ chlorophyll index (CI-G),³⁸ vegetation decay index (PSRI),³⁹ ratio vegetation stress index 2 (R2),⁴⁰ chlorophyll index (CI-RE), red edge vegetation stress index (RVSI),⁴¹ modified red edge ratio vegetation index (mSR₇₀₅),⁴² structure-insensitive pigment index (SIPI),⁴³ plant pigment ratio (PPR),⁴⁴ pigment-specific simple ratio 1 (PSSR1),⁴⁵ pigment-specific simple ratio 2 (PSSR2),⁴⁶ and red edge vegetation index ratio (VOG1).

Texture characteristics

In tree species classification, texture features play a critical role by capturing tonal variations within local spatial regions of imagery. These features effectively discriminate tree species with similar spectral signatures but distinct canopy structures, thereby enhancing classification accuracy. The Gray-Level Co-occurrence Matrix (GLCM) is a widely employed method for characterizing texture features by analyzing the spatial correlation properties of grayscale. This approach effectively captures the spatial characteristics of hyperspectral images.⁴⁷ As GLCM requires subsequent statistical computation to derive texture features, this study extracted eight second-order metrics: Second Moment, Contrast, Correlation, Entropy, Homogeneity, Mean, Dissimilarity, and Variance,⁴⁸ as shown in Table 3.

Given that principal component analysis (PCA) maximizes the retention of texture features during dimensionality reduction, this study selected the first principal component (which accounts for approximately 86.5% of the variance)—containing the most information in the hyperspectral imagery—to compute its Gray-Level Co-occurrence Matrix (GLCM). Texture features were extracted using second-order statistical measures with a grayscale quantization level of 64. Calculations were performed across five window sizes: 3×3, 5×5, 7×7, 9×9, and 15×15 pixels. Larger windows reduce image noise but increase edge blurring, whereas smaller windows enhance edge clarity and texture granularity at the cost of higher noise sensitivity. To optimize window selection, maximum likelihood classification (MLC) was applied to texture features from each window size. The 5×5 window yielding the highest classification accuracy was subsequently adopted for final classification.

Feature factors selection

This study integrates multiple feature factors—selected characteristic bands, vegetation indices, and texture feature factors—as inputs for the classification model to accurately distinguish tree species. Feature factors selection critically influences species identification efficacy; however, redundant feature factors may degrade model performance and stability.⁴⁹ To mitigate this, feature optimization through irrelevant feature factors elimination is essential. Recursive Feature Elimination with Cross-Validation (RFECV)⁵⁰

was employed for feature selection optimization. This method recursively removes the least contributive features while evaluating subset performance via cross-validation at each iteration, thereby enhancing model generalizability. The fundamental principles of RFECV are as follows:

- (1) Initial model fitting: The algorithm first fits the model across the entire set of feature factors. Performance metrics of the model, such as accuracy, mean squared error, or other relevant metrics, are recorded.
- (2) Feature factor importance ranking: After the initial fitting, the algorithm ranks the feature factors based on their importance. The specific metrics used for ranking depend on the selected machine learning algorithm.
- (3) Feature Factor Removal: Remove the least important feature factors from the feature factor set.
- (4) Model Re-fitting: Re-fit the model using the reduced feature factors. Record the model performance metrics again.
- (5) Iteration: Repeat steps 2 to 4 until the desired number of feature factors is reached or the model performance stabilizes.

As shown in [Figure 4](#), this study used RFECV to select feature factors and determine the feature subset that most effectively enhances overall model accuracy, includes 15 characteristic band factors, 7 vegetation index features, and 5 texture features. The vegetation index feature factors include PSR1, PSSR1, PSSR2, R2, CI-RE, ACI, and VOG1. The texture feature factors include Contrast, Homogeneity, Mean, Dissimilarity, and Variance.

Sample dataset creation and sample augmentation

This study employs a sliding window method to extract image blocks from hyperspectral imagery, dividing them into a series of spectral-spatial cubes centered on the geometric center pixel of each sample. Each cube has dimensions $S \times S \times L$ (where $S \times S$ represents the spatial neighborhood size and L denotes the number of spectral bands). The window size is set to 15×15 pixels to preserve sufficient spatial context while avoiding loss of detail from excessively large blocks. To balance spatial continuity of tree canopies and prevent information omission, the sliding step size was set to 10 pixels. This strategy ensures spatial independence of samples while fully capturing canopy edges, avoiding detail loss from excessive step sizes or sample redundancy from overly small steps. Subsequently, hierarchical random sampling was employed to partition the data into training, validation, and test sets, maintaining consistent distribution of all categories across subsets. During the partitioning process, each category adhered to a fixed 3:1:1 ratio, with the training set used for model training, the validation set employed to monitor and tune model performance, and the final model—selected based on the highest accuracy on the validation set—being evaluated on the independent test set. To meet sample requirements, this study necessitated a large-scale training dataset. Given the difficulty in acquiring field validation data, sample augmentation emerged as a common strategy to address this challenge. Geometric transformations, the most prevalent and effective image data augmentation method, were extensively applied throughout this study. Samples were rotated by 90° , 180° , and 270° around their center pixel, quadrupling the original sample count. Data augmentation was applied only to the training set, while the validation and test sets remained unaugmented to ensure accurate model performance evaluation. Additionally, to maintain sample balance, under-sampling is applied to the classes with a larger number of samples, while over-sampling is applied to the classes with fewer samples, as shown in [Table 4](#).

Random Forest algorithm

Random Forest (RF) is an ensemble machine learning methods widely employed for classification and regression tasks. Constructed upon decision trees, it enhances model performance and robustness through two key randomness mechanisms: feature randomization and bootstrap aggregation. Unlike single decision trees, RF's introduction of random feature subsets at each node split and random data sampling during tree construction significantly improves classification accuracy by reducing overfitting and increasing model diversity.

Support Vector Machine

Support Vector Machines (SVM) are a supervised classification method based on statistical learning theory, known for robust performance on complex and noisy data. SVM construct an optimal hyperplane that maximizes the margin between classes, defined by support vectors. As generalized linear classifiers, they perform binary classification by solving for the maximum-margin hyperplane via convex quadratic programming. SVM excel in small-sample learning and handle tasks such as classification and regression. Although fundamentally a binary classifier, SVM can be extended to multi-class problems by employing strategies such as the one-versus-one approach. For any two classes of samples, an SVM is designed, so for n classes of samples, $n(n-1)/2$ SVMs need to be designed. When classifying an unidentified sample, it is assigned to the category that secures the majority of votes.

2D-CNN

This study employs a 2D-CNN for hyperspectral tree species classification to address the limitation of traditional methods that rely solely on spectral information while ignoring spatial context, treating hyperspectral data as multi-channel images and jointly extracting spatial and spectral features through end-to-end training. The 2D-CNN architecture comprises nine layers: three convolutional layers, a max-pooling layer, a global average pooling layer, three fully connected layers, and a Dropout layer. Max-pooling reduces spatial dimensions while preserving prominent features, and global average pooling compresses the final feature maps into compact one-dimensional vectors. These vectors are then fed into the first two fully connected layers, which perform high-level feature

integration and nonlinear transformation, while the Dropout layer randomly masks neurons during training to mitigate overfitting. The final layer of the network is a fully connected layer with the number of neurons equal to the number of tree species categories. Its output is converted into a probability distribution for tree species classification via the Softmax function, thereby completing the identification of tree species for the input image patches.

QUANTIFICATION AND STATISTICAL ANALYSIS

Classification accuracy reflects the proportion of correctly classified pixels. Common accuracy assessment methods include the confusion matrix and the ROC curve; this study uses the confusion matrix. A confusion matrix is a contingency table where columns represent predicted classes (column sums indicate total predicted pixels per class) and rows represent true classes (row sums indicate total actual pixels per class). Based on the confusion matrix, the following metrics were calculated:

Overall accuracy (OA)

OA is simply the ratio of the correctly classified pixels to the grand total of pixels in the image.

$$OA = \frac{\sum_{i=1}^k C_{ii}}{N} \quad (\text{Equation 5})$$

Among them C_{ii} the diagonal element of the error matrix (i.e., the number of correctly classified pixels), N is the total number of pixels, and k is the number of categories.

Producer accuracy (PA)

PA refers to the ratio of the number of pixels correctly classified by the classifier to the total number of pixels in a given class, corresponding to the misclassification error.

$$PA = \frac{C_{ii}}{\sum_{j=1}^k C_{ij}} \quad (\text{Equation 6})$$

where C_{ii} is the number of correct classifications for class i , and $\sum_{j=1}^k C_{ij}$ is the total number of reference pixels for class i .

User accuracy (UA)

UA refers to the ratio of the total number of pixels correctly classified into a certain category to the total number of pixels classified into that category by the classifier, corresponding to the misclassification error.

$$UA = \frac{C_{ii}}{\sum_{j=1}^k C_{ji}} \quad (\text{Equation 7})$$

Where $\sum_{j=1}^k C_{ji}$ is the total number of pixels classified as class i .

Kappa coefficient

The Kappa coefficient is a comprehensive indicator for evaluating classification accuracy.

$$k = \frac{P_o - P_e}{1 - P_e} \quad (\text{Equation 8})$$

Where P_o is the observed probability of agreement and P_e is the expected probability of agreement.